



# Material Safety Data Sheet

HEBEI VEYONG BIO-CHEMICAL CO., LTD.

## MSDS FOR GLUFOSINATE-AMMONIUM 150g/L SL

### 1. PRODUCT IDENTIFICATION

Product Name: Glufosinate-ammonium 150g/L SL  
Active Ingredient: Glufosinate-ammonium, content 150g/L, CAS No.: 77182-82-2  
Chemical Name: Ammonium 4-[hydroxy(methyl)phosphinoyl]-DL-homoalaninate

### 2. COMPOSITION/INFORMATION ON HAZARDOUS INGREDIENTS

| Composition          | CAS No.    | Content (% by weight)   |
|----------------------|------------|-------------------------|
| Glufosinate-ammonium | 77182-82-2 | 13.5% (equal to 150g/L) |
| Others               |            | To 100%                 |

### 3. HAZARDS IDENTIFICATION

#### Emergency Overview

Warning! Causes substantial but temporary eye injury. Harmful if swallowed, inhaled or absorbed through the skin. Avoid contact with skin, eyes and clothing. Avoid inhalation of vapour or mist.

#### Physical State

liquid

#### Odor

weakly pungent

#### Appearance

Bluish homogeneous liquid

#### Routes of Exposure

Eye contact, Inhalation, Skin Absorption, Ingestion

#### Immediate Effects

##### Eye

Causes substantial but temporary eye injury. Severe eye irritation. Do not get in eyes.

##### Skin

Harmful if absorbed through skin. May cause slight irritation. Avoid contact with skin and clothing.

##### Ingestion

Harmful if swallowed. Do not take internally.

##### Inhalation

Harmful if inhaled. Avoid inhalation of vapour or mist.

##### Chronic or Delayed Long-Term

This product or its components may have target organ effects.

#### **4. FIRST AID MEASURES**

**General** When possible, have the product container or label with you when calling a poison control center or doctor or going for treatment.

**Eye** Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a physician or poison control center immediately.

**Skin**

Take off contaminated clothing and shoes immediately. Wash off immediately with plenty of water for at least 15 minutes. Call a physician or poison control center immediately.

**Ingestion**

Call a physician or poison control center immediately. Rinse out mouth and give water in small sips to drink. DO NOT induce vomiting unless directed to do so by a physician or poison control center. Never give anything by mouth to an unconscious person. Do not leave victim unattended.

**Inhalation**

Move to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a physician or poison control center immediately.

**Notes to Physician Signs and Symptoms**

Gastrointestinal disturbance, Tremors, Convulsions, Drowsiness, Diarrhoea, Vomiting, Nausea, muscular weakness, tachycardia, respiratory failure, unconsciousness, coma.

Symptoms may be delayed.

**Treatment**

Endotracheal intubation and gastric lavage, followed by administration of charcoal and sodium sulphate.

Appropriate supportive and symptomatic treatment as indicated by the patient's condition is recommended

#### **5. FIRE FIGHTING MEASURES**

**Flash point**

> 100 °C

**Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>), Foam, Water spray, Dry powder

**Fire Fighting Instructions**

Fight fire from upwind position. Keep out of smoke. Remove product from areas of fire, or otherwise cool containers with water in order to avoid pressure being built up due to heat. Whenever possible, contain fire-fighting water by diking area with sand or earth. Do not allow run-off from fire fighting to enter drains or water courses.

Firefighters should wear NIOSH approved self-contained breathing apparatus and full protective clothing.

## ***6. ACCIDENTAL RELEASE MEASURES***

### **Personal Precautions**

Isolate hazard area. Keep unauthorized people away. Avoid contact with spilled product or contaminated surfaces.

### **Methods for Cleaning Up**

Take up with absorbent material (e.g. sand, diatomaceous earth or a proprietary absorbent material). Collect and transfer the product into a properly labelled and tightly closed container. Clean contaminated floors and objects thoroughly, observing environmental regulations.

### **Additional Advice**

Use personal protective equipment. Do not allow to enter soil, waterways or waste water canal. Do not allow product to contact vegetation.

## ***7. HANDLING AND STORAGE***

### **Handling Procedures**

Handle and open container in a manner as to prevent spillage. Maintain exposure levels below the exposure limit through the use of general and local exhaust ventilation.

Keep away from heat and sources of ignition.

### **Storing Procedures**

Store in a cool, dry place and in such a manner as to prevent cross contamination with other pesticides, fertilizers, food, and feed. Store in original container and out of the reach of children, preferably in a locked storage area.

### **Work/Hygienic Procedures**

Wash hands thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, using the toilet or applying cosmetics.

Remove contaminated clothing immediately and dispose of safely.

Remove Personal Protective Equipment (PPE) immediately after handling this product. Before removing gloves clean them with soap and water. Wash thoroughly and put on clean clothing.

## ***8. EXPOSURE CONTROLS/PERSONAL PROTECTION***

### **General Protection**

Follow all label instructions. Train employees in safe use of the product. Eye wash facility and safety shower should be available.

Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and warm/tepid water. Keep and wash PPE separately from other laundry.

### **Eye/Face Protection**

Tightly fitting safety goggles

### **Hand Protection**

Chemical-resistant gloves (barrier laminate, butyl rubber, nitrile rubber or Viton)

### **Body Protection**

Wear long-sleeved shirt and long pants and shoes plus socks.

### **Respiratory Protection**

When respirators are required, select NIOSH approved equipment based on actual or potential airborne concentrations and in accordance with the appropriate regulatory standards and/or industry recommendations.

## ***9. PHYSICAL AND CHEMICAL PROPERTIES***

### **Appearance**

Bluish homogeneous liquid

### **Physical State**

liquid

### **Odor**

weakly pungent

### **pH**

4.0- 7.0

### **Density**

approx. 1.1 g/cm<sup>3</sup> at 20 °C

### **Water solubility**

soluble

## ***10. STABILITY AND REACTIVITY***

### **Incompatibility**

Bases

### **Hazardous Decomposition Products**

ammonia

### **Hazardous Reactions**

Stable under normal conditions.

## ***11. TOXICOLOGICAL INFORMATION***

Only acute toxicity studies have been performed on this product as formulated. The non-acute information pertains to the technical-grade active ingredient, glufosinate-ammonium.

### **Acute Oral Toxicity**

female combined rat: LD50:> 2000 mg/kg

### **Acute Dermal Toxicity**

rat: LD50: > 2,000 mg/kg

### **Acute Inhalation Toxicity**

male/female combined rat: LC50 (4h) : >4.82mg/l

### **Skin Irritation**

rabbit: Moderate skin irritation.

### **Eye Irritation**

rabbit: Moderate eye irritation.

**Sensitization**

guinea pig: Non-sensitizing.

**Chronic Toxicity**

Glufosinate-ammonium caused kidney effects and/or mortality at high dose levels in chronic toxicity studies in rats, mice and dogs.

**Assessment Carcinogenicity**

Glufosinate-ammonium was not carcinogenic in lifetime feeding studies in rats and mice.

**Reproductive & Developmental Toxicity**

REPRODUCTION: Glufosinate-ammonium was not a primary reproductive toxicant in rats. There was a decrease in the number of viable pups at the high dose in conjunction with maternal toxicity.

DEVELOPMENTAL TOXICITY: Glufosinate-ammonium was not a primary developmental toxicant in rats and rabbits. Developmental effects (e.g., delayed ossifications) were observed in rats but were considered secondary to maternal toxicity.

**Neurotoxicity**

Glufosinate-ammonium was not a neurotoxicant in acute and subacute neurotoxicity screening studies in rats. In a developmental neurotoxicity study in rats, offspring exposed to glufosinate-ammonium showed increases in locomotor activities. These effects occurred at high doses and in conjunction with maternal toxicity.

**Mutagenicity**

Glufosinate-ammonium was not mutagenic or genotoxic in a battery of in vitro and in vivo tests.

**12. ECOLOGICAL INFORMATION**

Glufosinate-ammonium is practically non-toxic to birds, fish, honeybees, earthworms and beneficial insects. It is practically non-toxic to Daphnia, but moderately toxic to sensitive aquatic species. It is expected to be slightly to moderately toxic to wild animals. DO NOT contaminate streams, rivers or waterways.

**Ecotoxicity**

Glufosinate-ammonium:

Fish toxicity: LC50 (96 h) rainbow trout 710 mg/L

LC50 (96 h) carp, bluegill sunfish, golden orfe > 1000 mg/L

Bird toxicity: Eight day dietary LC50 Japanese quail > 5000 mg/kg

Daphnia toxicity: Daphnia magna EC50 (48 h) 560 – 1000 mg/L

Algal toxicity: LD50 for Scenedesmus subspicatus  $\geq$  1000 mg/L

LD50 for Scenedesmus capricornutum 37 mg/L

**Environmental fate, persistence and degradability**

Aquatic, air or soil environmental hazards:

Glufosinate-ammonium is very soluble in water and is hydrolytically and photolytically stable. It is rapidly degraded in surface levels of soils and in water. This product is considered to be readily biodegradable. The potential for groundwater contamination with glufosinate-ammonium is minimal. Do not allow product to enter wastewater, rivers or creeks. Glufosinate-ammonium does not accumulate in the fatty tissues of fish or other animals.

### ***13. DISPOSAL CONSIDERATIONS***

#### **General Disposal Guidance**

Pesticide, spray mixture or rinse water that cannot be used according to label instructions may be disposed of on site or at an approved waste disposal facility.

#### **Container Disposal**

Do not use containers for other products. Triple rinse containers. Puncture container to avoid re-use.

Dispose of empty container in a sanitary landfill or by incineration, or, if allowed by State/Provincial and local authorities, by burning. If burned, stay out of smoke.

### ***14. TRANSPORT INFORMATION***

Non-hazardous for air, sea and road freight.

### ***15. REGULATORY INFORMATION***

#### **APPROVED HANDLER:**

The product shall be under the personal control of an approved handler when:

- a) the product is applied in a wide dispersive manner; or
- b) used by a commercial contractor.

However, it may be handled by a person who is not an approved handler if:

- an approved handler is present at the facility where the substance is being handled; and
- the approved handler has provided guidance to the person in respect to the handling of the product; and
- the approved handler is available at all times to provide assistance if necessary.

### ***16. OTHER INFORMATION***

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The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.